

~ 75

$$\gamma \sim R[0, 2\theta] \quad \mu\gamma = \frac{0+2\theta}{2} = \frac{3}{2}\theta \quad D\gamma = \frac{(2\theta-0)^2}{12} = \frac{\theta^2}{12}$$

а) Оценка методом моментов: $\mu\tilde{\theta}_1 = \frac{1}{n} \sum_{i=1}^n \tilde{\theta}_i = \frac{3}{2}\theta$

Оценка методом макс. правдоподобия: $l(\theta) = \prod_{i=1}^n p(x_i, \theta) = \frac{1}{\theta^n}$

$$x_{\min} \geq 0, \quad x_{\max} \leq 2\theta \Rightarrow \frac{x_{\max}}{2} \leq \theta \leq x_{\min} \\ \Rightarrow \sup l(\theta) = l\left(\frac{x_{\max}}{2}\right) \Rightarrow \hat{\theta}_2 = \frac{x_{\max}}{2}$$

б) Проверка на несмещенность:

$$\mu\tilde{\theta}_1 = \mu\left[\frac{2}{3} \frac{1}{n} \sum_{i=1}^n x_i\right] = \frac{2}{3} \mu\gamma = \frac{2}{3} \cdot \frac{3}{2}\theta = \theta \Rightarrow \text{Оценка Несмещенная}$$

$$\mu\tilde{\theta}_2 = \mu\left[\frac{x_{\max}}{2}\right] = \frac{1}{2} \mu(x_{\max}) \quad P(x_{\max} \leq y) = P(y) = F(y)$$

$$\varphi(y) = n F^{n-1}(y) p(y) = \frac{n}{\theta} \cdot \left(\frac{y-\theta}{\theta}\right)^{n-2}$$

$$\Rightarrow \mu\tilde{\theta}_2 = \frac{1}{2} \int_{-\infty}^{\infty} y \frac{n}{\theta^n} (y-\theta)^{n-2} dy = \frac{n}{2\theta^n} \int_{\theta}^{2\theta} y(y-\theta)^{n-2} dy =$$

$$\stackrel{(y-\theta=t)}{=} \frac{1}{2} \frac{n}{\theta^n} \int_0^{\theta} (t+\theta) t^{n-2} dt = \frac{1}{2} \frac{n}{\theta^n} \left(\frac{\theta^{n+1}}{n+1} + \frac{\theta^{n+2}}{n} \right) = \frac{1}{2} \frac{\theta n(n+1)}{n(n+1)} =$$

$$= \frac{\theta(2n+1)}{2n+2}$$

Смещение $\Rightarrow$ Несмещенная оценка

$$\tilde{\theta}_2' = \frac{2n+2}{2n+1} \cdot \frac{x_{\max}}{2}$$

Будет Несмещенная

Проверка состоятельности

$$D\tilde{\theta}_1 = \left(\frac{2}{3}\right)^2 \frac{1}{n^2} n D\gamma = \frac{4}{9n} \frac{\theta^2}{12} = \frac{\theta^2}{27n} \xrightarrow{n \rightarrow \infty} 0 \Rightarrow \text{Оценка Состоятельная}$$

$$\mu\tilde{\theta}_2' = \frac{1}{4} \mu x_{\max}^2 = \frac{1}{4} \int_{\theta}^{2\theta} y^2 (y-\theta)^{n-1} \frac{n}{\theta^n} dy = \frac{1}{4} \int_0^{\theta} (t+\theta)^2 \frac{n}{\theta^n} t^{n-1} dt =$$

$$= \frac{n}{4\theta^n} \int_0^{\theta} (t^{n+2} + 2\theta t^{n+1} + \theta^2 t^{n-1}) dt = \frac{1}{4} \frac{n}{\theta^n} \left(\frac{\theta^{n+3}}{n+3} + \frac{2\theta^{n+2}}{n+1} + \frac{\theta^{n+2}}{n} \right) =$$

$$= \frac{1}{4} \theta^2 \frac{n(n^2+n+2n+3n+n^2+2n+n+2)}{n(n+1)(n+2)} = \frac{\theta^2}{4} \frac{2n^2+4n+1}{(n+1)(n+2)}$$

$$\Rightarrow D\hat{\theta}_2^2 = \mu \hat{\theta}_2^2 - (\mu \hat{\theta}_2)^2 = \frac{1}{2} \theta^2 \frac{7n^2+4n+1}{(n+1)(n+2)} - \theta^2 \frac{4n^2+4n+1}{4(n+2)^2} =$$

$$= \theta^2 \left(\frac{4n^3+8n^2+9n+1}{4(n+1)^2(n+2)} - \frac{4n^3+4n^2+1}{4(n+2)^2} \right) =$$

$$= \theta^2 \frac{n}{4(n+1)^2(n+2)} \leftarrow D\hat{\theta}_2$$

$$\hat{\theta}_2' = \frac{2n+2}{2n+2} \hat{\theta}_2 \Rightarrow 2 \hat{\theta}_2 = \left(\frac{2n+2}{2n+2}\right)^2 D\hat{\theta}_2$$

$$= \left(\frac{2n+2}{2n+2}\right)^2 \frac{n}{4(n+1)^2(n+2)} \theta^2 \xrightarrow{n \rightarrow \infty} 0 \Rightarrow \hat{\theta}_2' \text{ satisfies the consistency condition}$$

c) - Проверка эффективности.

$$D\hat{\theta}_2 = \frac{\theta^2}{27n} ; D\hat{\theta}_2' = \left(\frac{2n+2}{2n+2}\right)^2 \frac{n}{4(n+1)^2(n+2)} \theta^2 = \frac{n}{(2n+2)^2(n+2)} \theta^2$$

$$\frac{\theta^2}{27n} \times \frac{n}{(2n+2)^2(n+2)} \theta^2 \Rightarrow \frac{(2n+2)^2(n+2)}{4n^3+12n^2+9n+2} \dots 27n^2$$

$$4n^3 - 15n^2 + 9n + 2 \geq 0 \Rightarrow n \geq 3 \Rightarrow D\hat{\theta}_2' < D\hat{\theta}_2 \Rightarrow$$

$\Rightarrow \hat{\theta}_2'$ более эффективен (при $n \geq 3$)

d) - Проверка гипотезы.

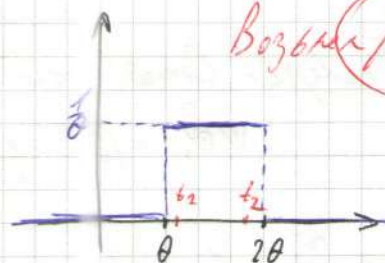
$$t_1 = q_{1-\beta} = \theta + \frac{\theta}{20} \cdot \frac{1}{2} = \frac{41}{40} \theta$$

$$t_2 = q_{\frac{\beta}{2}} = 2\theta - \frac{\theta}{40} = \frac{79}{40} \theta$$

$$F(x) = \frac{x-\theta}{\theta} ; \text{ тогда } y = \frac{x_{\max}}{\theta} \Rightarrow y \in [1; 2]$$

$$\Rightarrow P(y \leq y) = (y-1)^n \quad y \in [1; 2] \Rightarrow P(t_1 \leq y \leq t_2) = \beta$$

$$f(\bar{x}_n, \theta) = \frac{x_{\max}}{\theta} = y_{\max} \sim \varphi(y) \quad \varphi(y) = n(y-1)^{n-1} ; f(\bar{x}_n, \theta) = \frac{x_{\max}}{\theta} \sim \varphi(y)$$



Возьмем $\beta = 0.95$

$$t_1 = q_{1-\beta} ; t_2 = q_{\frac{\beta}{2}}$$

$$\Rightarrow P\left(\frac{x_{\max}}{t_2} < \theta < \frac{x_{\max}}{t_1}\right) = \beta$$

$$\int_1^{t_2} (1/t) dt = 0.075 \Rightarrow f(t_2) - f(1) = 0.075$$

$$\Rightarrow t_2 = 1 + (0.075)^{-1/n}$$

$$\text{Уточн. формула } t_2 = (1 + (0.075)^{-1/n})$$

⇒ Получили довер. интервал: $\left(\frac{x_{\max}}{t_2}, \frac{x_{\max}}{t_1}\right)$

е) Асимптотический довер. интервал при параметре θ

$$\text{Возьмем оценку из ОМТ } \tilde{\theta}_2' = \frac{2n+2}{n+2} \cdot \frac{x_{\max}}{2}$$

$$\Rightarrow \int_{-\infty}^{+\infty} d\left(\frac{\partial F}{\partial \theta}\right) = \int_{-\infty}^{+\infty} d\left(\frac{-\theta - (1-\theta)}{\theta^2}\right) = - \int_{-\infty}^{+\infty} \frac{1}{\theta^2} dx + 0 = \frac{\partial}{\partial \theta} \int_{-\infty}^{+\infty} dF$$

А вот все, все, кажется не так. Получили $\Rightarrow$ она не является
сильно регулярной $\Rightarrow$ ОМТ не подходит

Возьмем $\tilde{\theta}_2$. Сб-образован, потому Вн. ур. преобразован ЦДТ

$$\frac{\tilde{\theta}_n - \theta}{\sqrt{\tilde{\theta}_n - \theta}} \tilde{\theta}_n \leadsto N(0, 1), \quad \tilde{\theta}_2 = g(\tilde{\theta}) = \frac{2}{3} \tilde{\theta}_2$$

$$\Rightarrow \tilde{\theta}_n \frac{g(\tilde{\theta}) - g(\theta)}{\sigma(\tilde{\theta})} \leadsto N(0, 1);$$

$$\sigma^2(\tilde{\theta}) = \nabla^T g(\theta) K \nabla g(\theta) = \frac{4}{9} (\tilde{\theta}_2 - \theta)^2$$

$$\Rightarrow \tilde{\theta}_n \frac{\tilde{\theta}_2 - \theta}{\frac{2}{3} \sqrt{\tilde{\theta}_2 - \theta}} \leadsto N(0, 1) \Rightarrow t_2 = u_{\beta/2}, \quad t_1 = u_{1-\beta/2}$$

$$t_1 < \tilde{\theta}_n \frac{\tilde{\theta}_2 - \theta}{\frac{2}{3} \sqrt{\tilde{\theta}_2 - \theta}} < t_2 \Rightarrow \left(\tilde{\theta}_2 - \frac{2}{3} t_2 \sqrt{\frac{\tilde{\theta}_2 - \theta}{\tilde{\theta}_n}} < \theta < \tilde{\theta}_2 - \frac{2}{3} t_1 \sqrt{\frac{\tilde{\theta}_2 - \theta}{\tilde{\theta}_n}} \right)$$

довер. интервал

Пункты f, g, h относятся к Коде

LVS

$$p(x) = \begin{cases} \frac{\theta-1}{x^\theta}, & x \geq 1 \\ 0, & x < 1 \end{cases} \quad \theta > 1$$

2) Оценки в методе Минс. градиенте

$$L(\theta) = \prod_{i=1}^n \left(\frac{\theta-1}{x_i^\theta} \right) = (\theta-1)^n \prod_{i=1}^n \frac{1}{x_i^\theta} \quad \text{or} \quad (\theta-1)^n \left(\prod_{i=1}^n \frac{1}{x_i} \right)^\theta \Rightarrow$$

$$\Rightarrow \ln L(\theta) = n \ln(\theta - 2) + \theta \ln \left(\prod_{i=1}^n \frac{1}{x_i} \right) \Rightarrow \frac{d}{d\theta} \Rightarrow$$

$$\Rightarrow \frac{d}{d\theta} \ln L(\theta) = \frac{n}{\theta-1} + \ln\left(\prod_{i=1}^n \frac{1}{x_i}\right) = 0 \Rightarrow$$

$$\Rightarrow \theta = 1 + \frac{n}{-\ln(\prod_{i=1}^n x_i)}$$

$$\Rightarrow \frac{d^2}{d\theta^2}(\ln L(\theta)) = -\frac{1}{(\theta-3)^2} < 0 \Rightarrow \text{FOUR MAX}$$

$$\Rightarrow \hat{\theta} = 1 - \frac{n}{\ln\left(\prod_{i=1}^n \frac{1}{x_i}\right)}$$

В) Добр. Усербен гле Меднама

$$\int_{-\infty}^{\text{med}} p(x) dx = \int_{-\infty}^{\text{med}} \frac{\theta-1}{x^{\theta}} dx = (\theta-1) \left(\frac{1}{(1-\theta) \text{med}^{\theta-1}} - \frac{1}{(1-\theta) 1} \right) =$$

$$= 1 - \frac{1}{\text{med}^{a-1}} = \frac{1}{2} \Rightarrow \text{med} = 2^{\frac{1}{a-1}}$$

Нужно проверить на сильную регулярность.
 где тогда $F(x, \theta)$ — гонимая функция, зависящая от x и θ .
 $(1; +\infty)$ и $\frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{+\infty} dF(x, \theta) = \int_{-\infty}^{+\infty} d \frac{\partial^2 F(x, \theta)}{\partial \theta^2}$

resp. $\text{re}(1, +\infty) \Rightarrow F(x, \theta)$ - resp. group

$$\frac{\partial^2 F}{\partial \theta^2} = \frac{\partial^2}{\partial \theta^2} \left(1 - \frac{1}{x^{\theta-1}} \right) = \frac{\partial}{\partial \theta} \left(-\frac{\partial}{\partial \theta} x^{-(\theta-1)} \right) = \frac{\partial}{\partial \theta} \left(x^{-(\theta-1)} \ln x \right) = \frac{\ln^2 x}{x^{\theta-1}}$$

$\text{resp. re}(1, +\infty)$

$\Rightarrow F(x, \theta)$ - belongs to exp. group! $\text{re}(1, +\infty)$

$$\left\{ \begin{aligned} \frac{\partial}{\partial \theta} \int_{-\infty}^{+\infty} dF &= 0; \quad \int_{-\infty}^{+\infty} d \left(\frac{\partial F}{\partial \theta} \right) = \int_1^{+\infty} \frac{\partial F}{\partial \theta} dx = \\ &= \int_1^{+\infty} \frac{x^\theta - (\theta-1)x^\theta \ln x}{x^{\theta-1}} dx = \int_1^{+\infty} \frac{1}{x^\theta} dx - (\theta-1) \int_1^{+\infty} \frac{\ln x}{x^\theta} dx \end{aligned} \right\}$$

$$\int_1^{+\infty} d \left(\frac{\partial F}{\partial \theta} \right) \cdot \frac{\ln x}{x^{\theta-1}} \Big|_1^{+\infty} = 0 - 0 \stackrel{\ln x|_{x=1}=0}{=} 0 \Rightarrow \frac{\partial}{\partial \theta} \int_{-\infty}^{+\infty} dF = \int_{-\infty}^{+\infty} d \left(\frac{\partial F}{\partial \theta} \right)$$

$\frac{\ln x}{x^{\theta-1}} \xrightarrow{x \rightarrow +\infty} 0$
 $\theta > 1$

Проблема гиперболического $\Gamma(\theta)$ при непрерывности

$$\Gamma(\theta) = \int_1^{+\infty} \left(\frac{\partial \ln p}{\partial \theta} \right)^2 p dx = \int_1^{+\infty} \left(\frac{1}{\theta-1} - \ln x \right)^2 \frac{\theta-2}{x^\theta} dx =$$

$$= (\theta-1) \int_1^{+\infty} \left(\frac{1}{(\theta-1)^2} - \frac{2 \ln x}{\theta-2} + \ln^2 x \right) \frac{1}{x^\theta} dx = \left\{ \begin{aligned} t \in [0, +\infty) \\ x = e^t \\ dx = e^t dt \end{aligned} \right\} =$$

$$= \int_0^{+\infty} \frac{1}{\theta-2} e^{-t(\theta-1)} dt - 2 \int_0^{+\infty} t e^{-t(\theta-1)} dt + (\theta-1) \int_0^{+\infty} t^2 e^{-t(\theta-1)} dt$$

$$\Gamma_1 = \frac{1}{(\theta-2)^2} \int_0^{+\infty} e^{-t(\theta-1)} d((\theta-2)t) = \frac{1}{(\theta-2)^2} \left(-e^{-t(\theta-1)} \right) \Big|_0^{+\infty} = \frac{1}{(\theta-2)^2}$$

$\lim_{t \rightarrow +\infty} e^{-t(\theta-1)} = 0$
 $\lim_{t \rightarrow 0} e^{-t(\theta-1)} = 1$

$$\begin{aligned} \Gamma_2 &= \int_0^{+\infty} t e^{-t(\theta-1)} d((\theta-2)t) = -\frac{1}{\theta-2} \int_0^{+\infty} t d(e^{-t(\theta-1)}) = \\ &= \frac{1}{\theta-2} \left(t e^{-t(\theta-1)} \Big|_0^{+\infty} + \int_0^{+\infty} e^{-t(\theta-1)} dt \right) = \frac{1}{(\theta-2)^2} \int_0^{+\infty} e^{-t(\theta-1)} d(t(\theta-1)) = \\ &= \frac{1}{(\theta-2)^2} \left(-e^{-t(\theta-1)} \right) \Big|_0^{+\infty} = \frac{1}{(\theta-2)^2} \end{aligned}$$

$\lim_{t \rightarrow +\infty} t e^{-t(\theta-1)} = 0$
 $\lim_{t \rightarrow 0} t e^{-t(\theta-1)} = 0$
 $\lim_{t \rightarrow +\infty} e^{-t(\theta-1)} = 0$
 $\lim_{t \rightarrow 0} e^{-t(\theta-1)} = 1$

$$I_3 = \int_0^{\infty} t^2 e^{-t(\theta-1)} d(t(\theta-1)) = -t^2 e^{-t(\theta-1)} \Big|_0^{\infty} + \int_0^{\infty} 2t e^{-t(\theta-1)} dt =$$

$\begin{matrix} \nearrow 0 \\ \text{if } \theta > 1 \\ \searrow 0 \\ \text{if } \theta = 0 \end{matrix}$

$$= 2 I_2 = \frac{2}{(\theta-1)^2}$$

Выраж: $I(\theta) = \frac{1}{(\theta-1)^2}$ — непрерывна на множестве $\mathcal{U}_3(1; +\infty)$

$\Rightarrow$ т.е. $I(\theta)$ — непрерывна на множестве $\mathcal{U}_3(1; +\infty)$

$$\hookrightarrow \frac{\partial}{\partial \theta} \int_{-\infty}^{+\infty} dF(x, \theta) = \int_{-\infty}^{+\infty} d \frac{\partial F}{\partial \theta}$$

$\Rightarrow$ Лемма Ризнера

$$\frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{+\infty} dF = 0; \quad \int_{-\infty}^{+\infty} d \left(\frac{\partial^2 F}{\partial \theta^2} \right) = \int_1^{\infty} d \left(\frac{\ln^2 x}{x^{\theta-1}} \right) = \frac{\ln^2 x}{x^{\theta-1}} \Big|_1^{\infty} = 0$$

$\begin{matrix} \nearrow 0 \\ \text{if } \theta > 1 \\ \searrow 0 \\ \text{if } \theta = 0 \end{matrix}$

$$\Rightarrow \frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{+\infty} dF = 0 = \int_{-\infty}^{+\infty} d \left(\frac{\partial^2 F}{\partial \theta^2} \right) \quad *$$

$\Rightarrow$ из непрерывности, следует непрерывность $F(x, \theta)$ на $(1; +\infty)$. $\square$

$\hookrightarrow *$ $\Rightarrow$ т.е.

Лемма Числова Ризнера

$$\Rightarrow \sqrt{n}(\tilde{\theta} - \theta) \rightsquigarrow \mathcal{N}\left(0, \frac{1}{I(\theta)}\right)$$

$$\text{med} = 2^{\frac{1}{\theta-1}} \text{ — группа; } \text{med}'(\theta) = 2^{\frac{1}{\theta-1}} \ln 2 \cdot \frac{1}{(\theta-1)^2} \Big|_{\theta=\theta} = -2^{\frac{1}{\theta-1}} \ln 2 \cdot \frac{1}{(\theta-1)^2} \neq 0$$

$\begin{matrix} \nearrow 0 \\ \text{if } \theta > 1 \end{matrix}$

$$\Rightarrow \text{Розенберга } \sqrt{n}(\text{med}(\tilde{\theta}) - \text{med}(\theta)) \rightsquigarrow \mathcal{N}\left(0, \text{med}'^2(\theta) \cdot \frac{1}{I(\theta)}\right)$$

$$\sigma(\theta) = 2^{\frac{1}{\theta-1}} \ln 2 \cdot \frac{1}{(\theta-1)^2} \cdot (\theta-1) = \frac{\ln 2 \cdot 2^{\frac{1}{\theta-1}}}{\theta-1}$$

$$\Rightarrow \sqrt{n} \frac{\text{med}(\tilde{\theta}) - \text{med}(\theta)}{\sigma(\tilde{\theta})} \rightsquigarrow \mathcal{N}(0, 1)$$

$$t_2 = \frac{U_{1+\beta}}{2} \quad t_2 = \frac{U_{1+\beta}}{2}$$

$$t_2 < \hat{\sigma} \left(\frac{\text{med}(\tilde{\theta}) - \text{med}(\theta)}{\hat{\sigma}(\tilde{\theta})} \right) < t_2 \Rightarrow$$

$$\Rightarrow \left(\text{med}(\tilde{\theta}) - \frac{\hat{\sigma}(\tilde{\theta})}{\hat{\sigma}} t_2 < \text{med}(\theta) < \text{med}(\tilde{\theta}) - \frac{\hat{\sigma}(\tilde{\theta})}{\hat{\sigma}} t_2 \right)$$

c) Асимптотически гауссовский дисперсион гнл θ

$$\sqrt{n}(\tilde{\theta} - \theta) \sim N(0; \frac{1}{I(\theta)}) \Rightarrow$$

$$\Rightarrow \sqrt{n}(\tilde{\theta} - \theta) \sqrt{I(\tilde{\theta})} \sim N(0, 1) \Rightarrow$$

$$\Rightarrow \tilde{\theta} - \frac{t_2}{\sqrt{n I(\tilde{\theta})}} < \theta < \tilde{\theta} - \frac{t_2}{\sqrt{n I(\tilde{\theta})}}$$

Пыраба d, t, f - Вугл.

Бере намы Бабэг 775. Лган $\theta = 17$

Ас. гнл med: (1.0395; 1.0599)

Ас. гнл θ : (12.546; 13.175)

Клс. Bootstrap: (12.992; 17.449)

Пар. Bootstrap: (12.029; 17.813)

$$L = 5.629$$

$$L = 5.157$$

$$L = 5.784$$

А гнл ме Бабэг кнл гнл 55

Потн. кнл (9.998; 10.181)

Ас. кнл (10.015; 10.523)

Клс. Bootstrap (9.889; 10.639)

$$L = 0.183$$

$$L = 0.508$$

$$L = 0.752$$